Assignment 1

MAKINI Caleb

2026-09-28

assignment

setting working directory

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

loading the libraries

```
library(dplyr)
library(rio)
library(ggplot2)
library(tidyverse)
library(psych)
library(formatR)
```

importing data

```
kl <- import("data1.csv")
head(kl)
```

```
  StudentID Age StudyHours SleepHours Attendance ExamScore PreviousGPA
1         1  20          5          7         95        85         3.5
2         2  22          3          6         80        72         3.2
3         3  19          6          8         92        88         3.7
4         4  21          4          5         75        68         3.0
5         5  20          7          9         98        92         3.9
6         6  23          2          6         70        65         2.8
  FamilyIncome
1        45000
2        52000
3        38000
4        61000
5        42000
6        55000
```

Data cleaning progress

Checking if their exist missing values

```
miss_val <- sum(is.na(kl))
miss_val
```

```
[1] 0
```

identification of missing varuables

```
colSums(is.na(kl))
```

```
   StudentID          Age   StudyHours   SleepHours   Attendance    ExamScore
           0            0            0            0            0            0
 PreviousGPA FamilyIncome
           0            0
```

Removing missing values

```
clean_data <- na.omit(kl)
head(clean_data)
```

```
  StudentID Age StudyHours SleepHours Attendance ExamScore PreviousGPA
1         1  20          5          7         95        85         3.5
2         2  22          3          6         80        72         3.2
3         3  19          6          8         92        88         3.7
4         4  21          4          5         75        68         3.0
5         5  20          7          9         98        92         3.9
6         6  23          2          6         70        65         2.8
  FamilyIncome
1        45000
2        52000
3        38000
4        61000
5        42000
6        55000
```

checking whether clean data still have any missing value

```
colSums(is.na(clean_data))
```

```
   StudentID          Age   StudyHours   SleepHours   Attendance    ExamScore 
           0            0            0            0            0            0 
 PreviousGPA FamilyIncome 
           0            0 
```

random reploacing of missing values with mean

Descriptive statistics

```
describe(kl)
```

```
             vars   n     mean      sd  median  trimmed      mad     min   max
StudentID       1 571   286.00  164.98   286.0   286.00   212.01     1.0   571
Age             2 571    20.69    1.38    20.0    20.61     1.48    19.0    23
StudyHours      3 571     4.75    1.64     5.0     4.75     1.48     2.0     8
SleepHours      4 571     6.85    1.42     7.0     6.88     1.48     4.0     9
Attendance      5 571    84.66    9.67    86.0    85.19    11.86    65.0    98
ExamScore       6 571    78.94   10.07    79.0    79.17    13.34    60.0    94
PreviousGPA     7 571     3.36    0.39     3.4     3.36     0.44     2.6     4
FamilyIncome    8 571 48075.31 8373.91 47000.0 47910.28 10378.20 35000.0 63000
                range  skew kurtosis     se
StudentID      570.0  0.00    -1.21   6.90
Age              4.0  0.33    -1.14   0.06
StudyHours       6.0  0.06    -0.82   0.07
SleepHours       5.0 -0.15    -0.89   0.06
Attendance      33.0 -0.32    -1.07   0.40
ExamScore       34.0 -0.08    -1.21   0.42
PreviousGPA      1.4 -0.04    -1.02   0.02
FamilyIncome 28000.0  0.13    -1.16 350.44
```

replace the missing values of age with mean

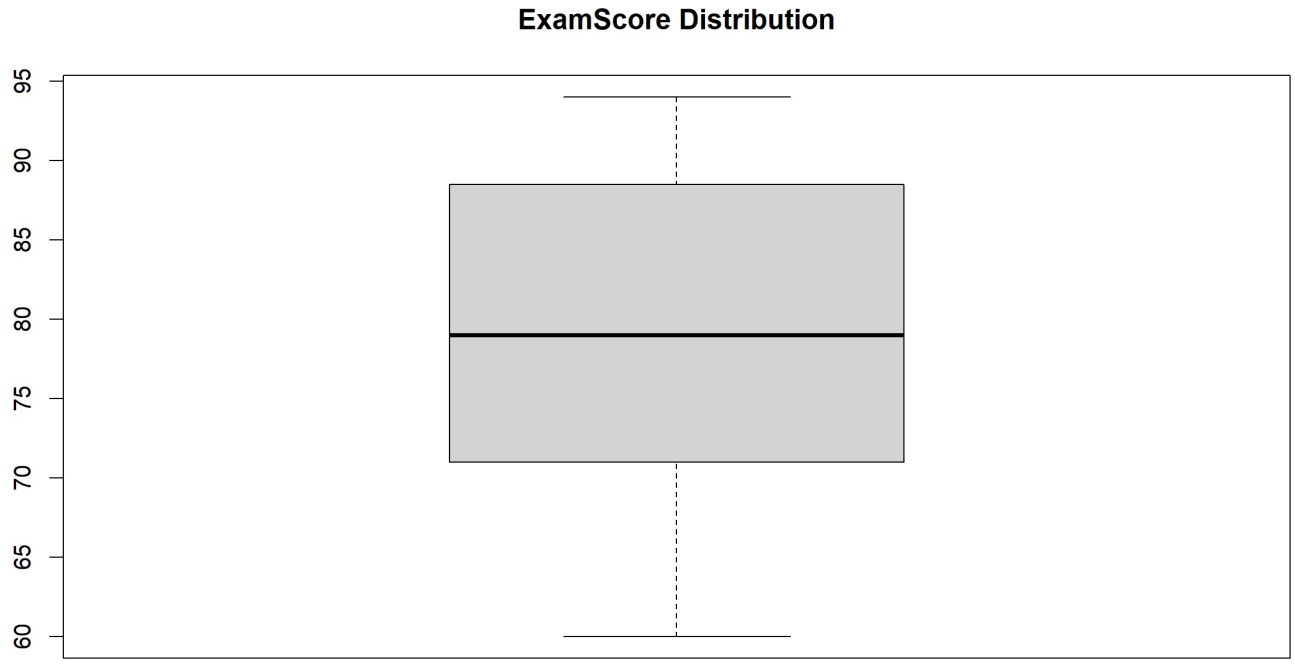
```
kl$Age[is.na(kl$Age)] <- mean(kl$Age, na.rm = TRUE)
```

conclusion

DAta above has no missing values

checking existence of outliers

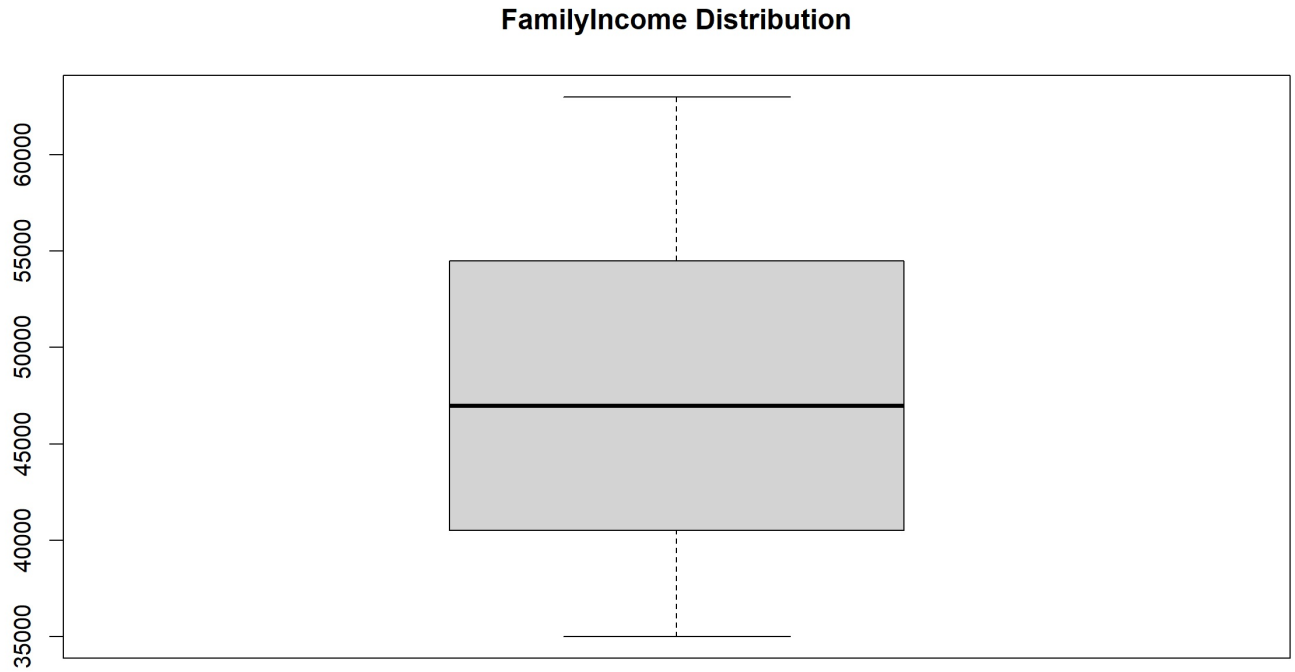
```
boxplot(kl$ExamScore, main = "ExamScore Distribution")
```



Exam score lacks outliers

checking using family Income

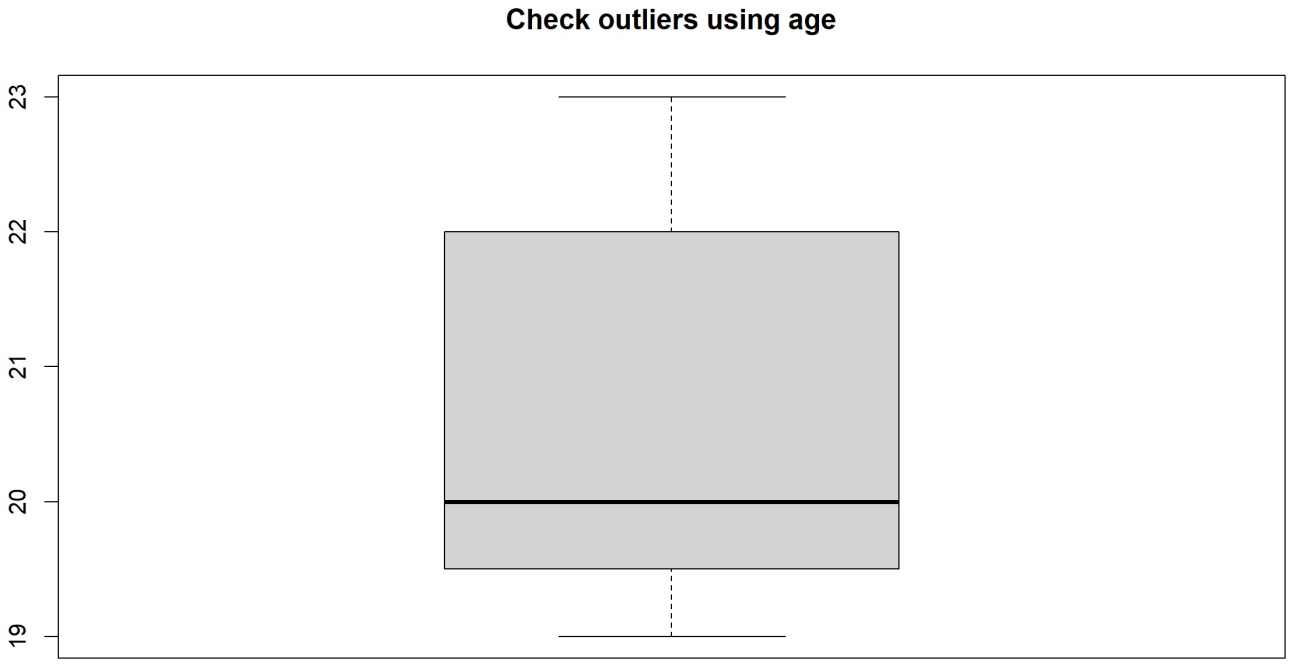
```
boxplot(kl$FamilyIncome, main = "FamilyIncome Distribution")
```



FamilyIncome Free from outliers

checking using Age

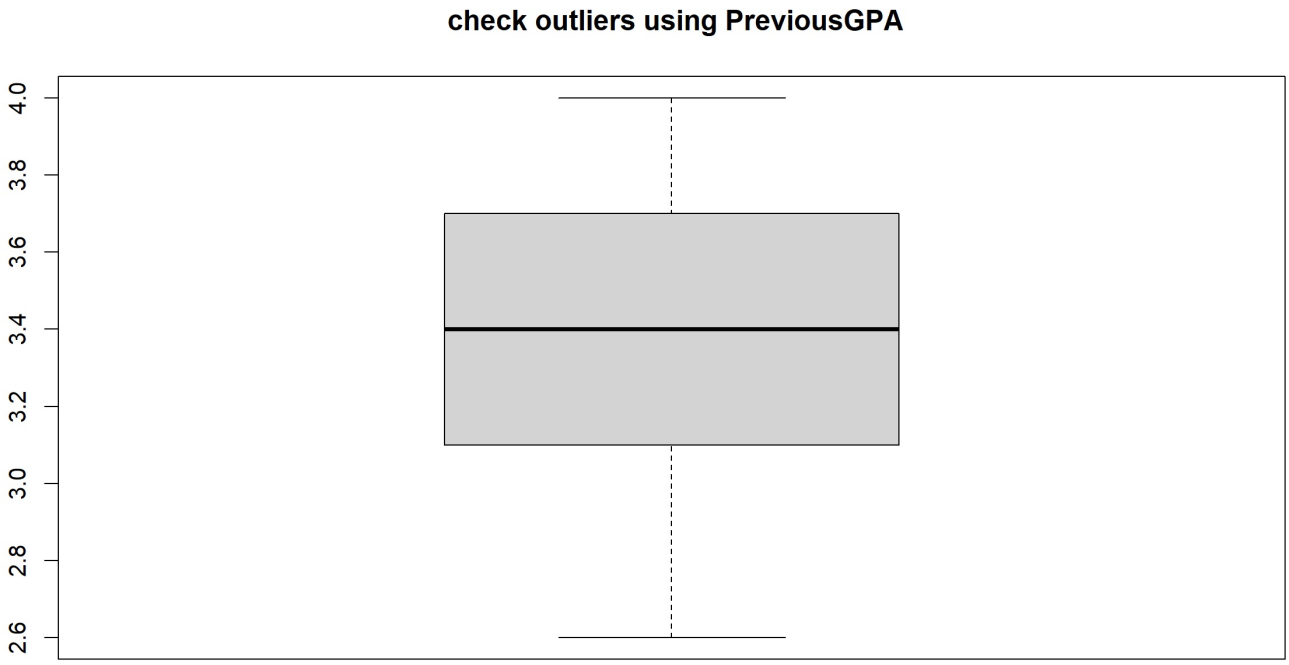
```
boxplot(kl$Age, main = "Check outliers using age")
```



Outliers not present while using age

Checking using PreviousGPA

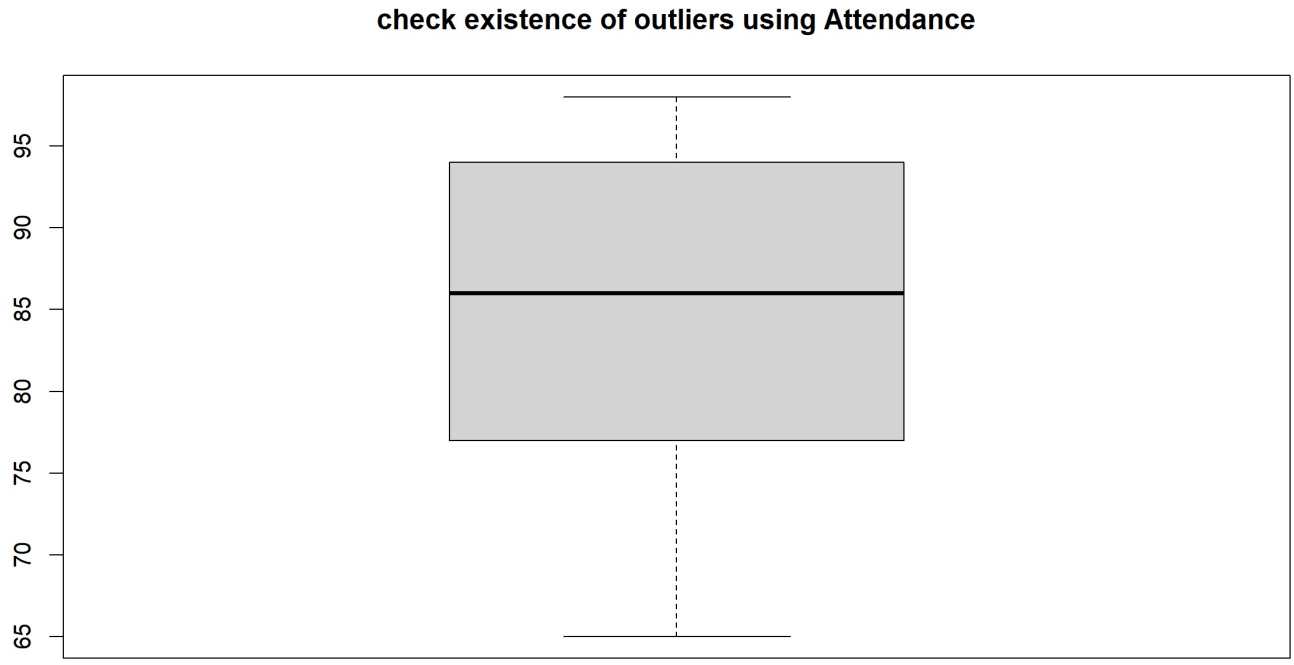
```
boxplot(kl$PreviousGPA, main = " check outliers using PreviousGPA")
```



Outliers unavailable using PreviousGPA

checking using Attendance

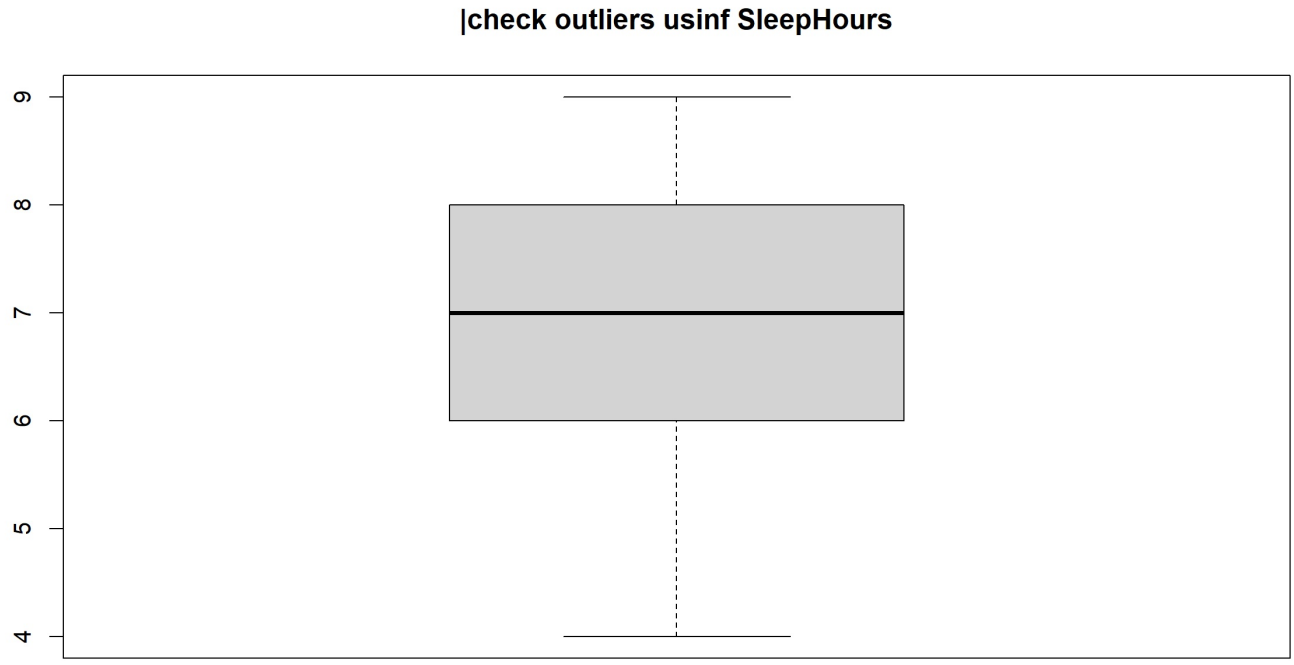
```
boxplot(kl$Attendance, main = "check existence of outliers using Attendance")
```



clean from outliuers using Attendance

checking using SleepHours

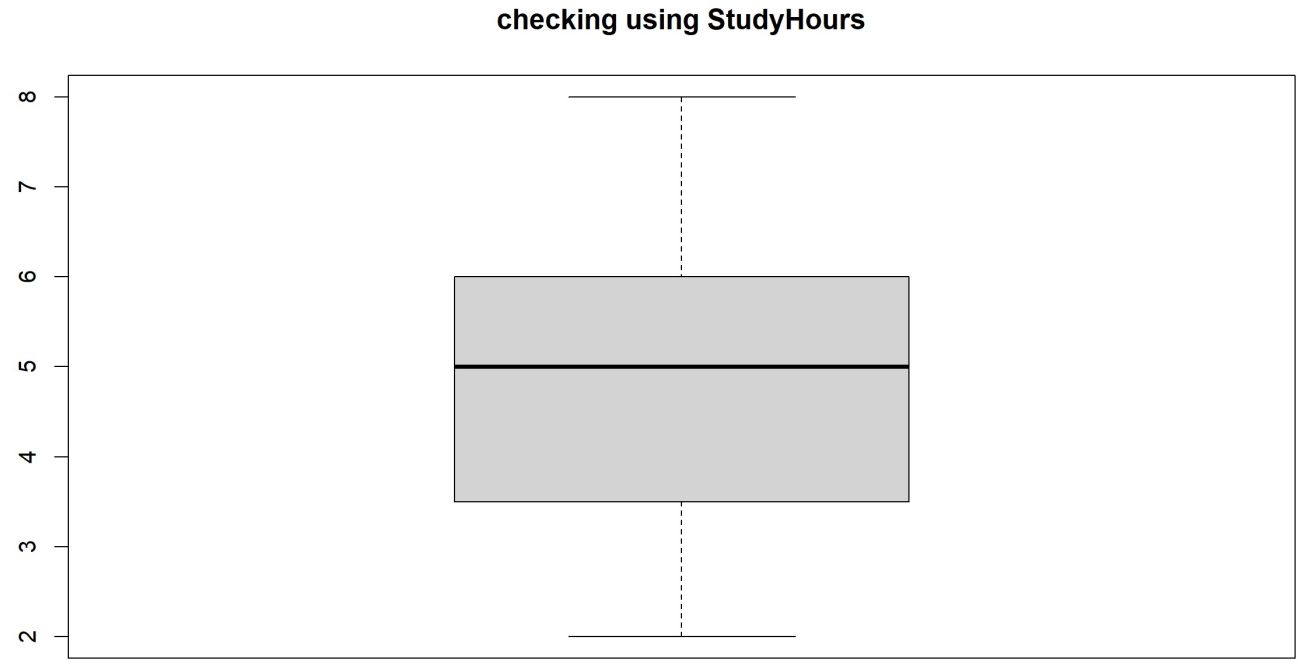
```
boxplot(kl$SleepHours, main = "|check outliers usinf SleepHours")
```



clean from outliuers using SleepHours

checking using StudyHours

```
boxplot(kl$StudyHours, main = "checking using StudyHours")
```



clean from outliuers using Study hours

assuming outliers exists. Then

we handle outliers using 1.5 IQR rule

```
Q1 <- quantile(kl$ExamScore, 0.25)
Q3 <- quantile(kl$ExamScore, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR
cp <- kl[kl$ExamScore >= lower_bound & kl$ExamScore <= upper_bound, ]
```

mean, median, varaince and standard deviation of the data

```
mean_val <- mean(cp$ExamScore, na.rm = TRUE)
median_val <- median(cp$ExamScore, na.rm = TRUE)
variance_val <- var(cp$ExamScore, na.rm = TRUE)
sd_val <- sd(cp$ExamScore, na.rm = TRUE)
print(paste("mean:", mean_val))
```

```
[1] "mean: 78.9404553415061"
```

```
print(paste("median:", median_val))
```

```
[1] "median: 79"
```

```
print(paste("variance:", variance_val))
```

```
[1] "variance: 101.438553476511"
```

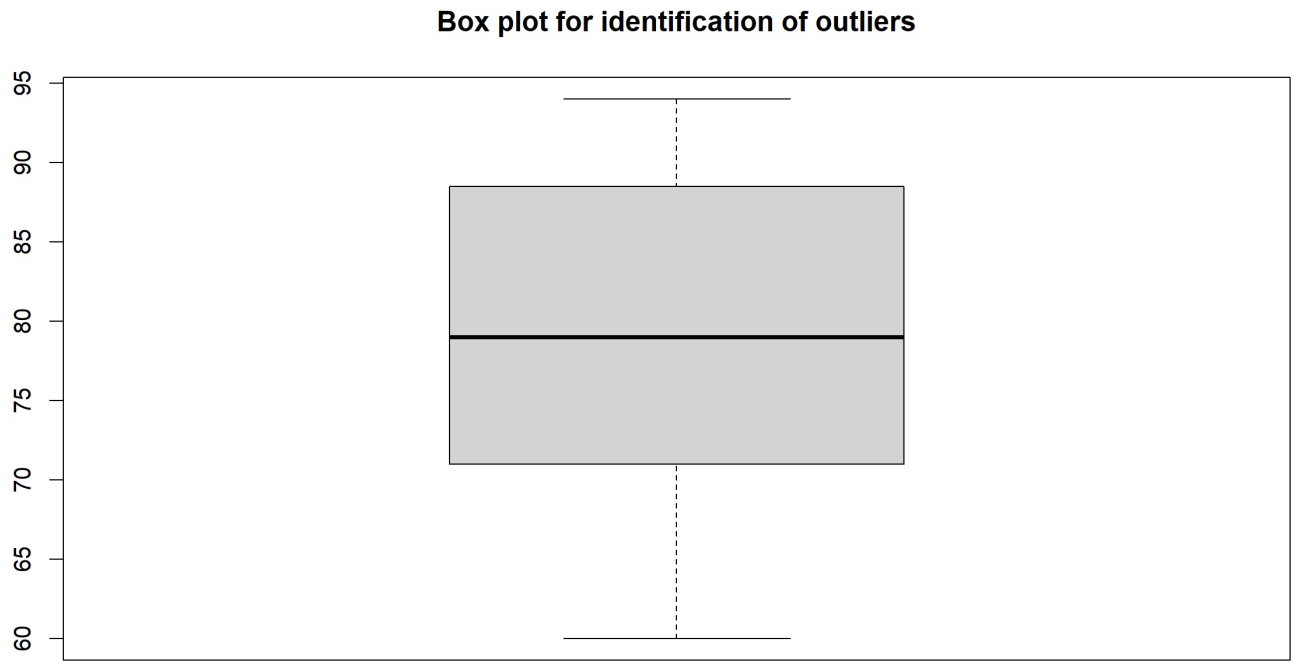
```
print(paste("standard deviationincome:", sd_val))
```

```
[1] "standard deviationincome: 10.0716708383719"
```

plotting cp cleaned data

Verification & identification of outliers

```
boxplot(cp$ExamScore, main = "Box plot for identification of outliers")
```

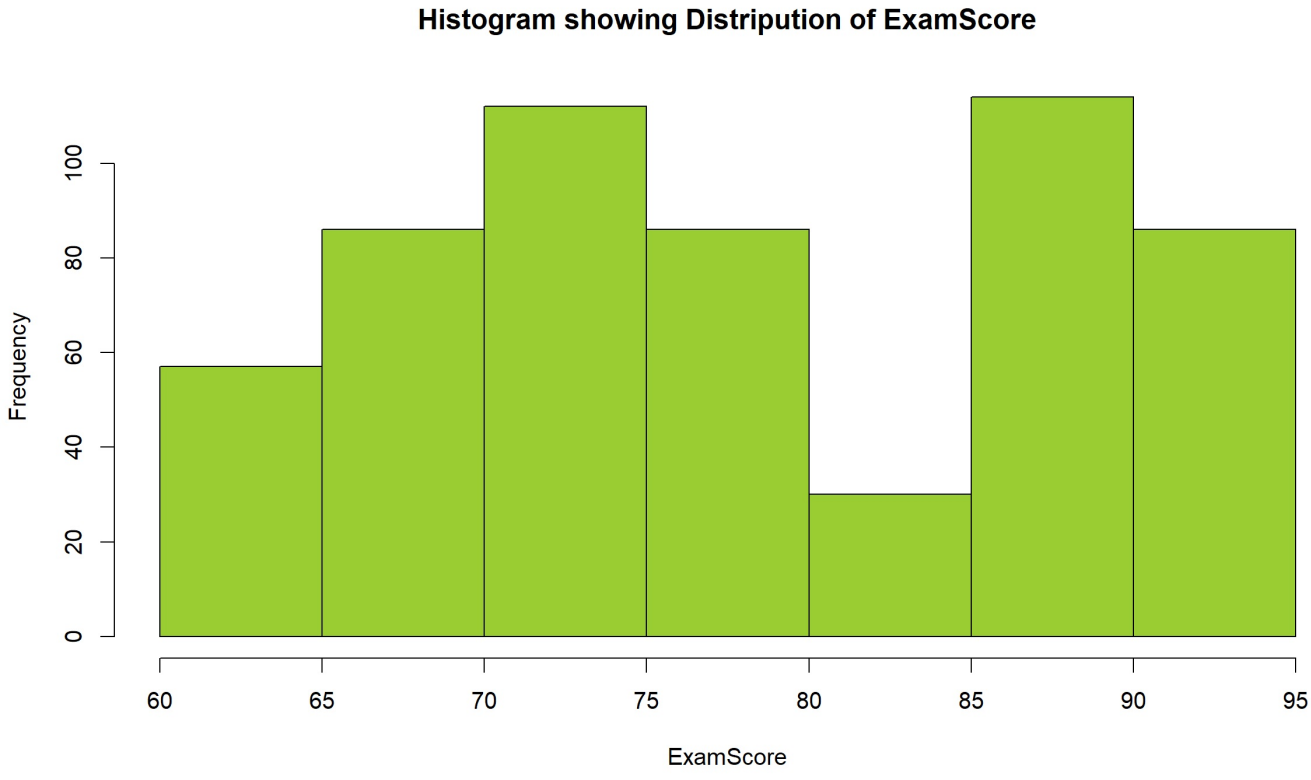


data free from outliers

Visualization using histogram

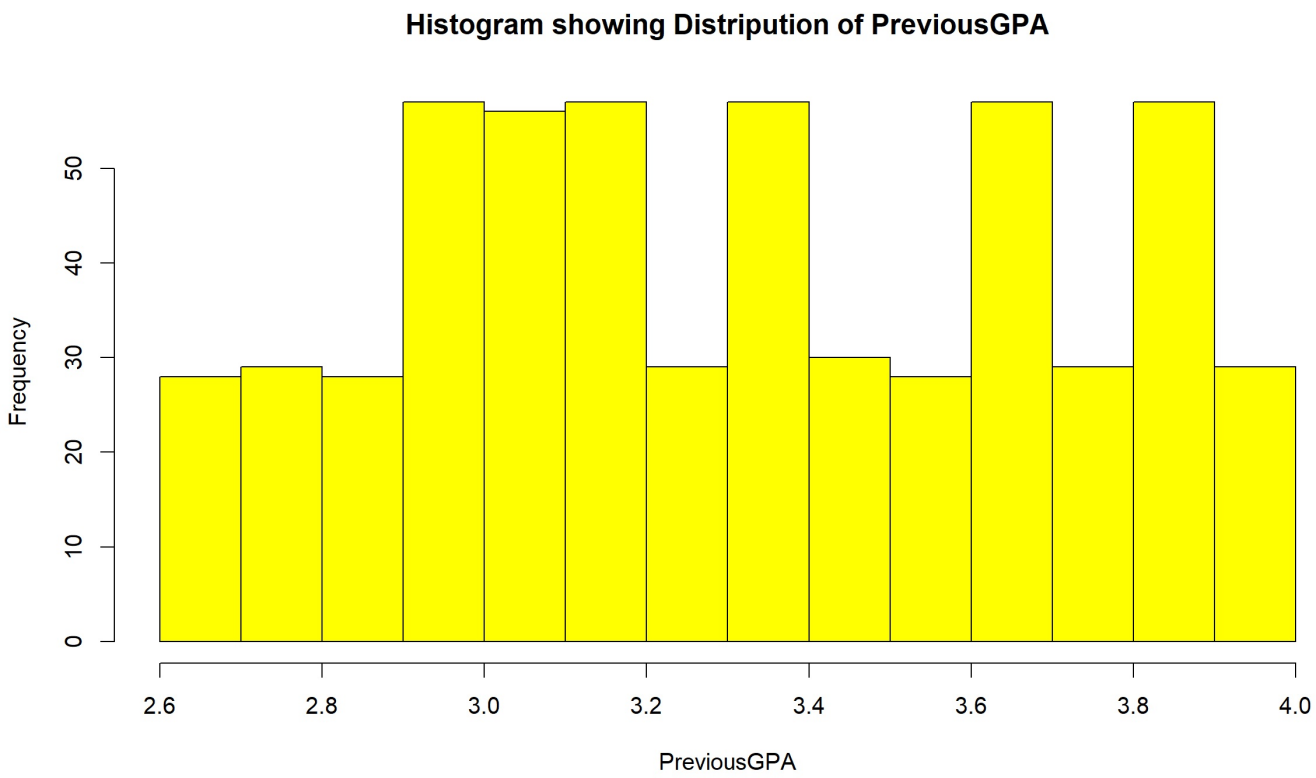
exam score

```
hist(cp$ExamScore, xlab = "ExamScore", ylab = "Frequency", main = "Histogram showing Distripution of ExamScore",
    col = "yellowgreen")
```



using previousGPA

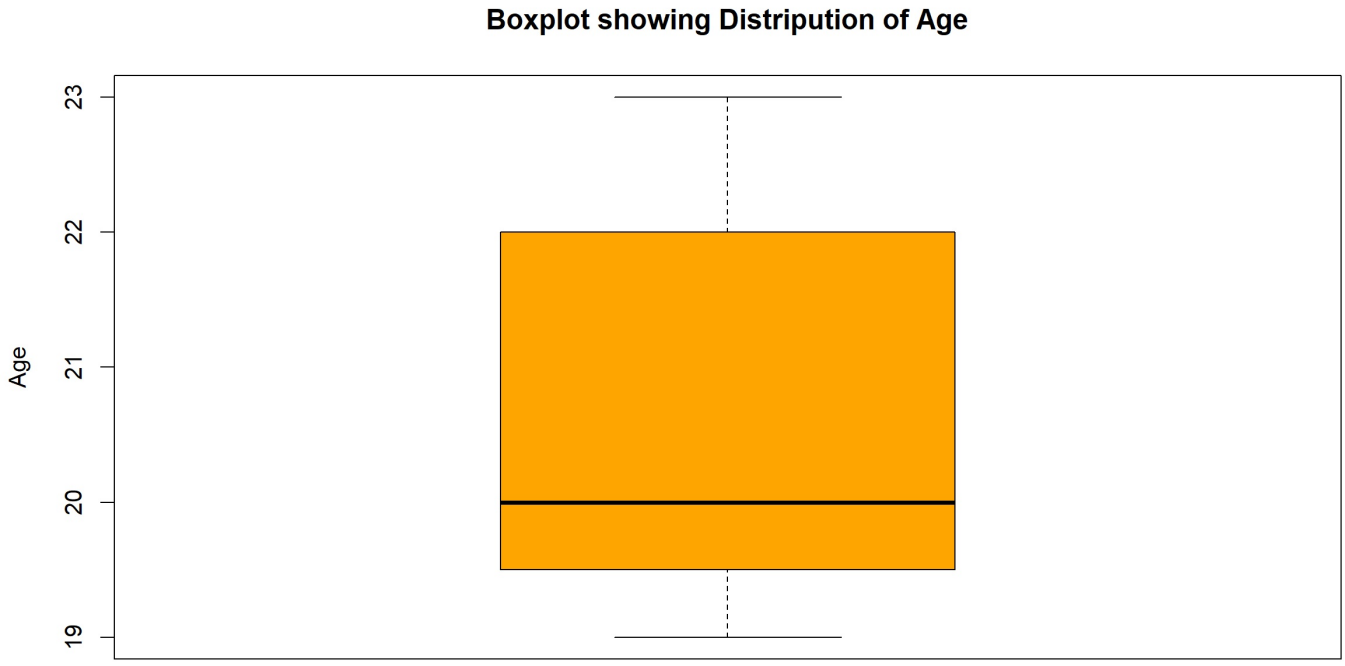
```
hist(cp$PreviousGPA, xlab = "PreviousGPA", ylab = "Frequency", main = "Histogram showing Distripution of Previous GPA",
    col = "yellow")
```



visualization using boxplot

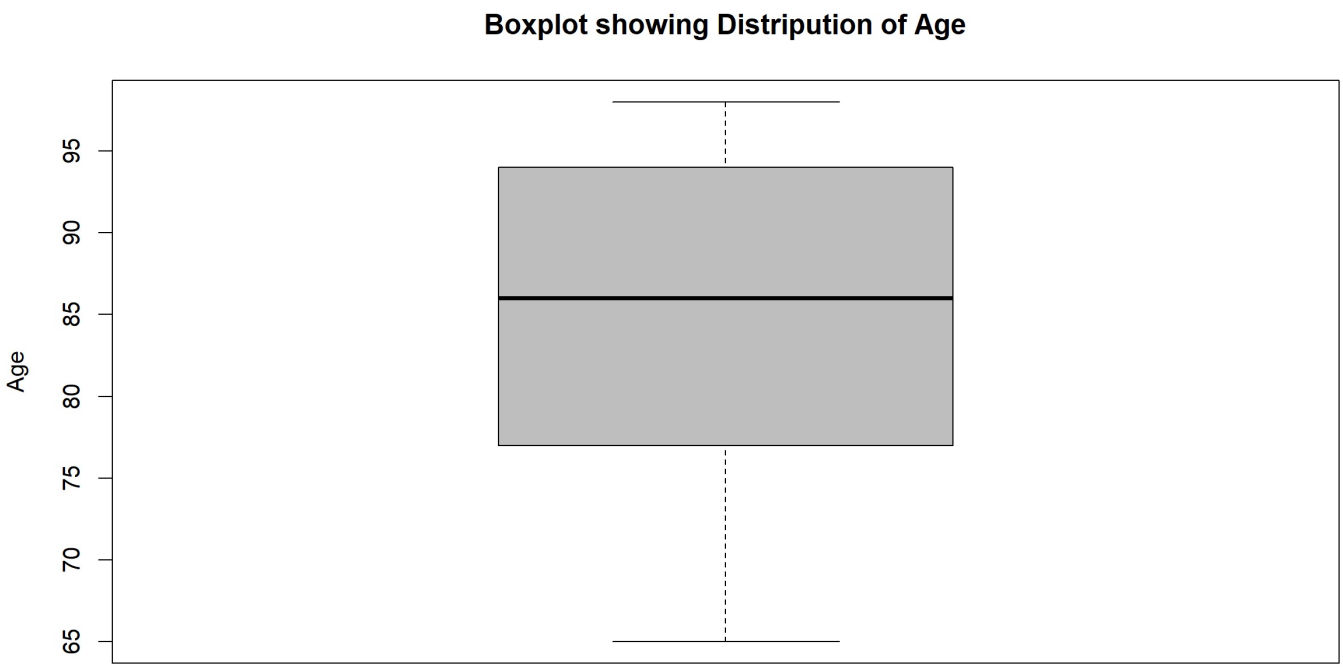
boxplot using Age

```
boxplot(cp$Age, ylab = "Age", main = "Boxplot showing Distripution of Age", col = "orange")
```



boxplot using Attendance

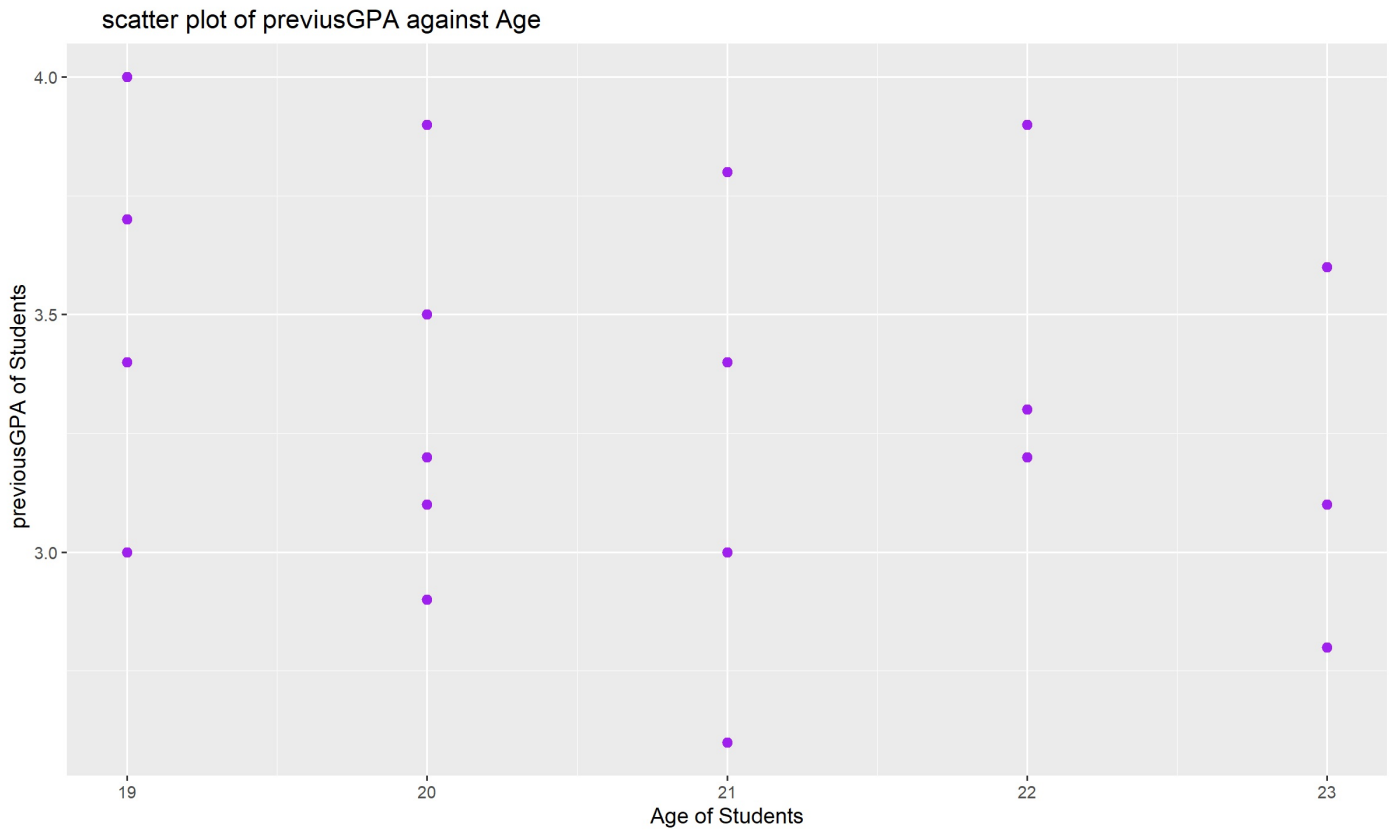
```
boxplot(cp$Attendance, ylab = "Age", main = "Boxplot showing Distripution of Age",
    col = "grey")
```



visualization using scatterplots

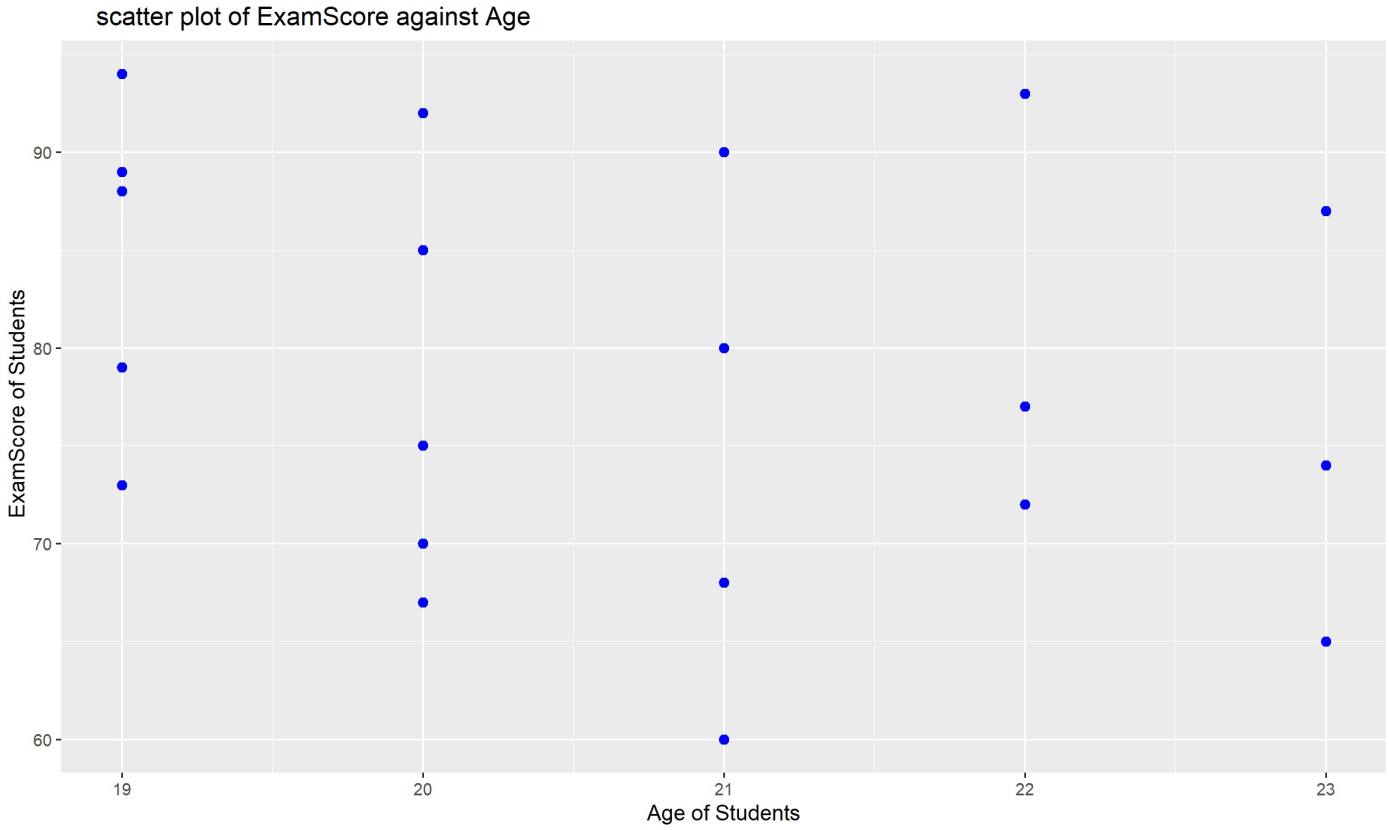
Scatter plot using previusGPA against Age

```
ggplot(cp, aes(x = Age, y = PreviousGPA)) + geom_point(col = "purple", size = 2) +
    labs(x = "Age of Students", y = " previousGPA of Students", title = "     scatter plot of previusGPA against
Age")
```



Scatter plot using ExamScore against Age

```
ggplot(cp, aes(x = Age, y = ExamScore)) + geom_point(col = "blue", size = 2) + labs(x = "Age of Students",
    y = " ExamScore of Students", title = "     scatter plot of ExamScore against Age")
```



Scatter plot using ExamScore against PreviousGPA

```
ggplot(cp, aes(x = PreviousGPA, y = ExamScore)) + geom_point(col = "brown", size = 2) +
    labs(x = "Age of Students", y = " ExamScore of Students", title = "     scatter plot of ExamScore against Pre
viousGPA")
```

